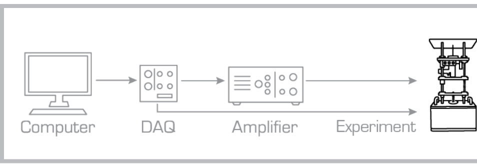
QUANSER

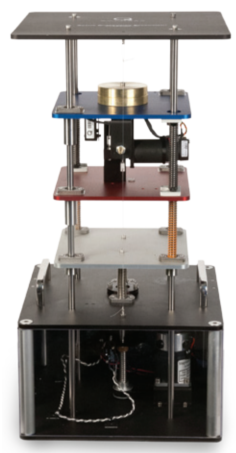
Quick Start Guide: Active Suspension

MATLAB SIMULINK

STEP 1 Check Components and Details

Make sure your Active Suspension experiment includes the following components:





1. Active Suspension
2. Set of three 5-pin DIN to 5-pin DIN encoder cables
3. RCA to RCA analog cable
4. Quanser Workstations Resources DVD*
   (includes controllers; digital versions of User Manual, Quick Start Guide and courseware; and other files)

*DVD supplied with the QUARC Real-Time Rapid Control Prototyping software, see Step 2

STEP 2 Additional Components Required for Set Up

To complete the Active Suspension experiment set up, you will also need the following:

1. QUARC Real-Time Rapid Control Prototyping software Installation DVD
2. Power Amplifier (AMPAQ-L2 pictured)
3. One of the following data acquisition devices:
   a. Quanser Q8-USB, or
   b. QPID/QPIDe with Terminal Board
4. Set of two RCA to RCA cables
5. Set of two 3-pin DIN to 4-pin DIN motor cables

Note: These components must be purchased separately.

To set up your Active Suspension experiment, please read the following instructions carefully.

STEP 3 Install and Test QUARC

A. Make sure you have all required software, as listed in the QUARC Compatibility Table document located in the QUARC DVD folder.

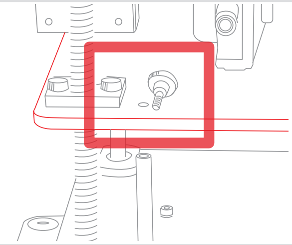
B. See the QUARC Installation Manual for details on how to install the software.

C. Make sure you test the system using the Sine and Scope demo. You can access this by typing qc_show_demos in the Matlab prompt.

STEP 4 Set Up the Hardware

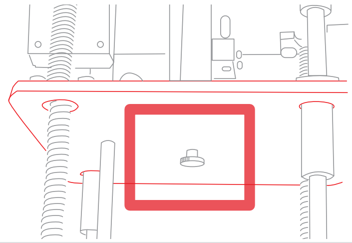
Follow the procedure below to set up your Active Suspension experiment. The connections shown below are illustrated using a generic data acquisition (DAQ) device and an AMPAQ-L2 amplifier (you may have a different DAQ or amplifier). For detailed instructions, see the Active Suspension experiment User Manual (enclosed with shipment).

A



Detach the red plate from the white plate by removing the thumb screw from the top of the red plate.

B



Detach the red plate from the blue plate by removing the thumb screw from the bottom of the red plate.

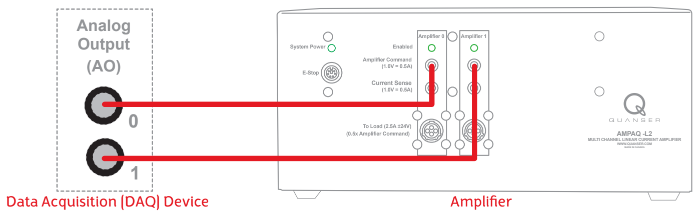
C

Before proceeding, your data acquisition (DAQ) device should be already set up and tested. For detailed instructions, see the DAQ Quick Start Guide or User Manual.

D

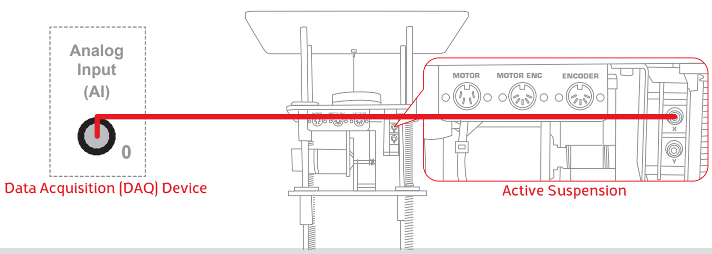
Make sure everything is powered OFF before making any of these connections. This includes turning off your PC and the amplifier.

E



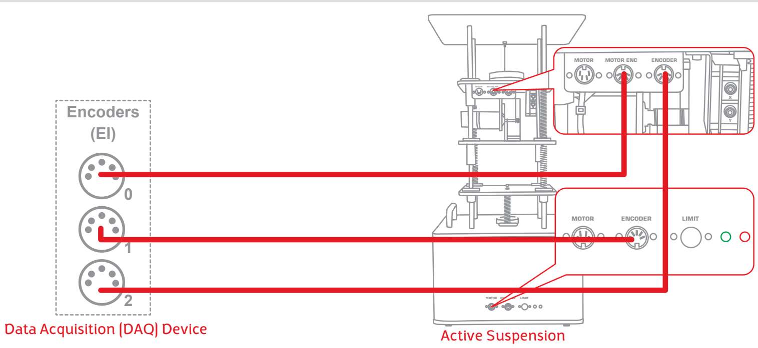
Using the RCA to RCA cable, connect the **Analog Output (AO) Channel #0** on the data acquisition (DAQ) device to the **Amplifier Command #0** connector on the amplifier. Then, connect **(AO) #1** on the DAQ device to the **Amplifier Command #1** connector on the amplifier.

F



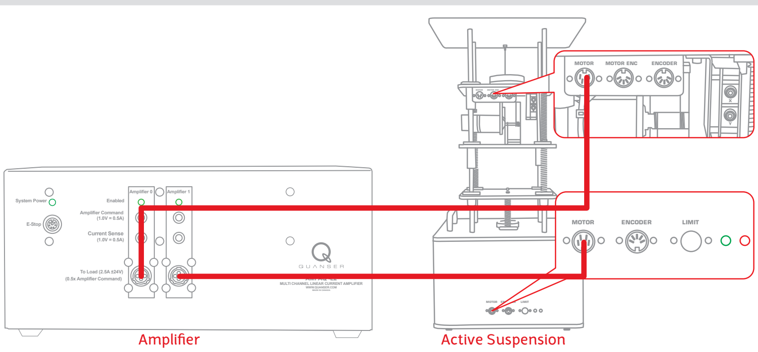
Using the RCA to RCA cable, connect **Analog Input (AI) Channel #0** on the data acquisition (DAQ) device to the **X** accelerometer connector on the Active Suspension.

G



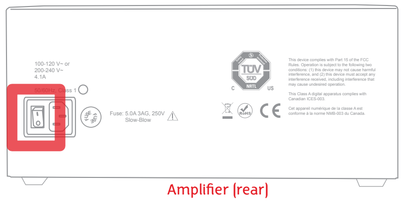
Using the 5-pin-DIN to 5-pin-DIN encoder cable, connect the **Encoder #0** (ENC #0) on the DAQ device to the **ENCODER** connector on the top panel of the Active Suspension. Similarly, connect the **ENC #1** and **ENC #2** on the DAQ device to the **ENCODER** connector on the bottom panel of the Active Suspension and the **MOTOR ENC** connector on the top panel of the Active Suspension.

H



Using the 3-pin-DIN to 4-pin-DIN motor cable, connect the **TO LOAD #0** on the amplifier to the **MOTOR** connector on the top panel of the Active Suspension. Also connect **TO LOAD #1** on the amplifier to the **MOTOR** connector on the bottom panel of the Active Suspension.

I



Amplifier (rear)

Turn ON the power switch on the AMPAQ-L2. It is located on the rear of the device.

STEP 5 Testing the Active Suspension Experiment

Go through the procedure below to test your Active Suspension experiment.

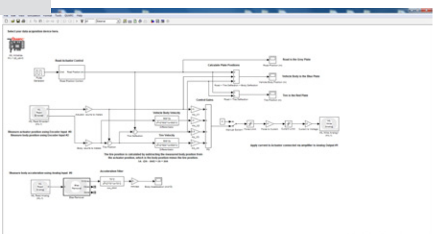
A

Make sure your PC and amplifier are powered ON.

B

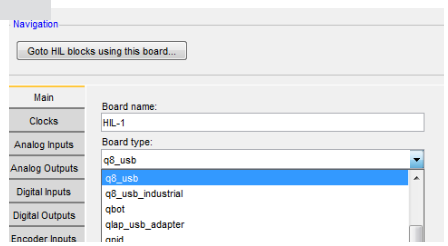
On the Resources DVD (supplied with the QUARC and Active Suspension package), locate the **Quick Start Folder**: Mechatronics\Active Suspension\Quick Start. Copy the Quick Start folder to your local hard drive.

C



Open the Simulink model file (.mdl) found under the Quick Start folder on your hard drive.

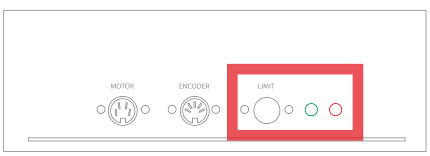
D



Configure HIL Initialize window

Double-click on the HIL Initialize block and choose the board that is installed on your system (e.g. Q8-USB).

E

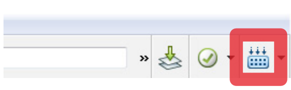


Active Suspension

Manually move the white plate until it hits the upper and lower limit switches, i.e. when the red LED is lit. Then, move the plate to the mid position and push the **Limit** button to turn on the green LED, i.e enable the motor power.

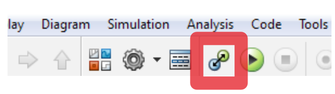
F

Click on the **Build Model** button on the Simulink model toolbar.



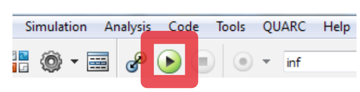
G

Once the model code has been compiled, click on the **Connect To Target** button.

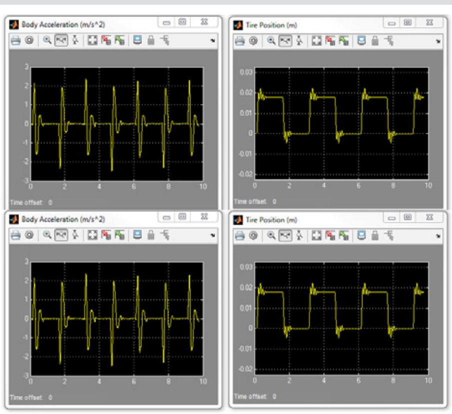


H

Click on the **Run** button to start the QUARC real-time model.



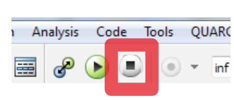
I



The scopes should look similar to those shown here. The white plate should be moving up and down to track the square wave. While the red and blue plate oscillations are minimized. If not, go to the *Troubleshooting* section.

J

Click on the Simulink Stop button to **stop** the running model.



TROUBLESHOOTING

Please review the following recommendations before contacting Quanser's technical support engineers.

1. Make sure the cables are firmly connected.
2. Refer to the Troubleshooting Guide in the Active Suspension User Manual.

Getting an error when trying to build or run the Quick Start Simulink model (.mdl)	**A.** Type ver in the Matlab Command Window and verify that QUARC is on the list. If not, then go through the QUARC Quick Installation Guide to install QUARC. If it is listed, run mex-setup as described in the QUARC Installation Guide. **B.** If the "... specific kernel level driver for the specified card could not be found" error is prompted when you attempt to run, then you may not have selected the correct data acquisition (DAQ) device in the HIL Initialize block or the DAQ device has not been installed properly (refer to the DAQ device User Manual).
The white plate is not moving.	**A.** Ensure the green LED on the bottom panel is on. If not, follow the instructions in Section H of Step 5. **B.** Review connection in Steps 4E, ,4H and I. Make sure cables are firmly connected. **C.** Ensure the power amplifier is working, i.e., when using AMPAQ-L2 verify that the green LED next to the power label is on. **D.** Ensure the power amplifier is getting enabled i,e., when using AMPAQ-L2 verify that the green LED under the **Channel 0** label turns on when the VI is started. **E.** Verify the data acquisition device is functional.
The Active Suspension controller is not working.	**A.** Review connection in Steps 4E, 4H, and 4I. **B.** Ensure the power amplifier is powered on and operational, i.e., when using AMPAQ-L2 verify that the green LED next to the power label is lit. **C.** Ensure the power amplifier is getting enabled i.e., when using AMPAQ-L2 verify that the green LED under the **Channel 1** label turns on when the VI is started. **D.** Verify the data acquisition device is functional.
The Encoder is not reading.	**A.** Review connections in Step 4G. **B.** Verify that the data acquisition (DAQ) device is functional. Go through the DAQ User Manual for troubleshooting guidelines.

STILL NEED HELP? For further assistance from a Quanser engineer, contact us at tech@quanser.com or call +1-905-940-3575.